

ARSHIA AKHAVAN

Master's Student in Computer Science

@letmemakenewone@gmail.com

619-609-3341

San Diego, CA

arshiaakhavan.me

arshia-akhavan

ArshiAAkhavan

0009-0007-9631-6964



EXPERIENCE

Software Engineer

Sotoon

Jan 2023 – Dec 2024

Tehran, Iran

Sotoon is a B2B cloud provider that focuses on reliable and scalable computing solutions. I worked in the K8s-as-a-Service team, designing and maintaining Sotoon's Kubernetes engine and several integrations to Sotoon's services, such as CSI, CNI, and Auto-scaler.

Key Achievements:

- Automate management of **90+** Kubernetes clusters, optimizing operational efficiency and reducing tickets by **30%**.
- Achieve a **40%** reduction in initial cluster **costs** by eliminating duplicated system components and introducing **multi-tenancy**.
- Redesign cluster network topology to enhance **security** and abstract control plane management from customers.
- Plan and Lead **Migration** of over **100 customers'** k8s-clusters to the new infrastructure with **zero downtime**.
- Implement A/B testing framework for cluster updates, preventing **5** otherwise catastrophic failures.
- Investigate and debug a critical **data race** issue in the CSI plug-in that could have resulted in the deletion of over **70TB** of client storage data, ensuring data integrity and service reliability.
- Automate CI/CD pipeline for OS image creation, reducing new k8s version release time **from 1 week to 1 day**.

Data Platform Engineer

Digikala

Aug 2021 – Jan 2023

Tehran, Iran

Digikala is the largest e-commerce company in Iran, serving over 40 million users. As part of the Big Data & AI team, I was responsible for building and maintaining the infrastructure required to efficiently store, process, and manage the vast amounts of data generated by user activity.

Key Achievements:

- Developed a **Parallel Data Query Engine** on top of **Apache Spark**, revising data retrieval models and queries to reduce **resource usage** by 23% and **computation time** by 43%.
- Reduced API latency** by identifying blocking operations in an asynchronous workflow, optimizing the critical path to **triple requests per second (RPS)**.
- Designed and maintained a data platform using **K8s** for orchestration, **Ceph** for storage, **Airflow** for scheduling, **Spark** for batch/stream processing, and **Kafka** with **Debezium** for database synchronization.
- Implemented **GitOps**-based infrastructure management with **ArgoCD**, automating API deployments via **Kustomize** and **Helm**, reducing release cycles from 6 months to 6 hours.
- Deployed and maintained a **staging environment** and built a framework for the Data Science team, reducing their development time significantly.

DevOps Engineer

Tapsell

Nov 2019 – Aug 2021

Tehran, Iran

Tapsell is a leading AdTech company in Iran. As part of the DevOps team, I provided cloud infrastructure and platform services that enabled other technical teams to build, deploy, and scale their applications efficiently.

Key Achievements:

- Optimizing Cassandra's **Disk usage** for different workloads, resulting in 90% reduction in disk usage for some of our clusters.
- Designed and implemented a fully automated backup, validation, and restore system for our cloud-native infrastructure, replacing the legacy solution while preserving **backward compatibility** to enable seamless, incremental migration of 30TB+ of data with zero downtime.
- Built and maintained infrastructure by deploying a staging cluster on **K8s**, developing CI/CD pipelines with **GitLab CI**, automating provisioning with **Terraform** on **VMWare**, and administering Linux-based VMs to resolve network bottlenecks and improve performance.

PUBLICATIONS

Journal Articles

- A. Akhavan, A. Hosseinpour, A. Heydarnoori, and M. Keshani, "Linkanchor: An autonomous llm-based agent for issue-to-commit link recovery," *arXiv preprint*, vol. arXiv:2508.12232, 2025, Submitted 17 August 2025; revised 2 September 2025. DOI: [10.48550/arXiv.2508.12232](https://doi.org/10.48550/arXiv.2508.12232).

VOLUNTEER EXPERIENCE

Technical Chapter

Student's Scientific Chapter (SSC)

Aug 2020 – Aug 2021 Tehran, Iran

Maintain Technical resources owned by SSC and supervise technical requirements of events held by SSC

Chief of Infrastructure and System Team

Sharif AI Challenge 2021

Dec 2020 – May 2021 Tehran, Iran

Develop and deploy a scalable infrastructure/software for AI challenge's high demanding game engine. [GitHub](#)

Chief Technical Staff

ICPC

May 2023 Tehran, Iran

Build contest-customized OS images and distribute them on contestant hosts using Network boot.
Manage and configure hosts used by contestants and monitor their connectivity.
Restrict external access and host-to-host access of each contestant's host device.

Chief Technical Staff

ICPC west championship

Dec 2019 Tehran, Iran

Manage and configure hosts used by contestants and monitor their connectivity

EDUCATION

M.Sc. in Computer Science

San Diego State University

Sep 2025 – Currently CA, USA

B.Sc. in Computer Engineering

Sharif University of Technology

Sep 2018 – Apr 2024 Iran

HONORS & AWARDS

BGSU ACM Hackathon 2'nd place

Apr 2025 Bowling Green, OH

Awarded for developing [Piqniq](#), an emergency alert app designed to support users during panic attacks.

PROGRAMMING LANGUAGES

Rust

Go

C

Bash

Python

Java

SQL

TECHNOLOGIES

parallel Computing

AVX

Cuda

OpenMP

rayon-rs

crossbeam-rs

Orchestration

k8s

Docker

Docker-swarm

Networking

Nginx

HAproxy

Calico

CI/CD

Ansible

terraform

GitlabCI

ArgoCD

Bigdata

Spark

Airflow

Database & Storage Systems

Ceph

minIO

Kafka

Redis

Cassandra

MongoDB

LANGUAGES

English

French

Persian